

# Using positional information for predicting transcription factor binding sites

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## Introduction

Transcription factors (TF) play a central role in the mechanism of transcription. **Those proteins bind DNA**, each TF usually recognising a specific set of k-mers. TFs often bind together with other TFs and **cooperate to regulate the transcription**.

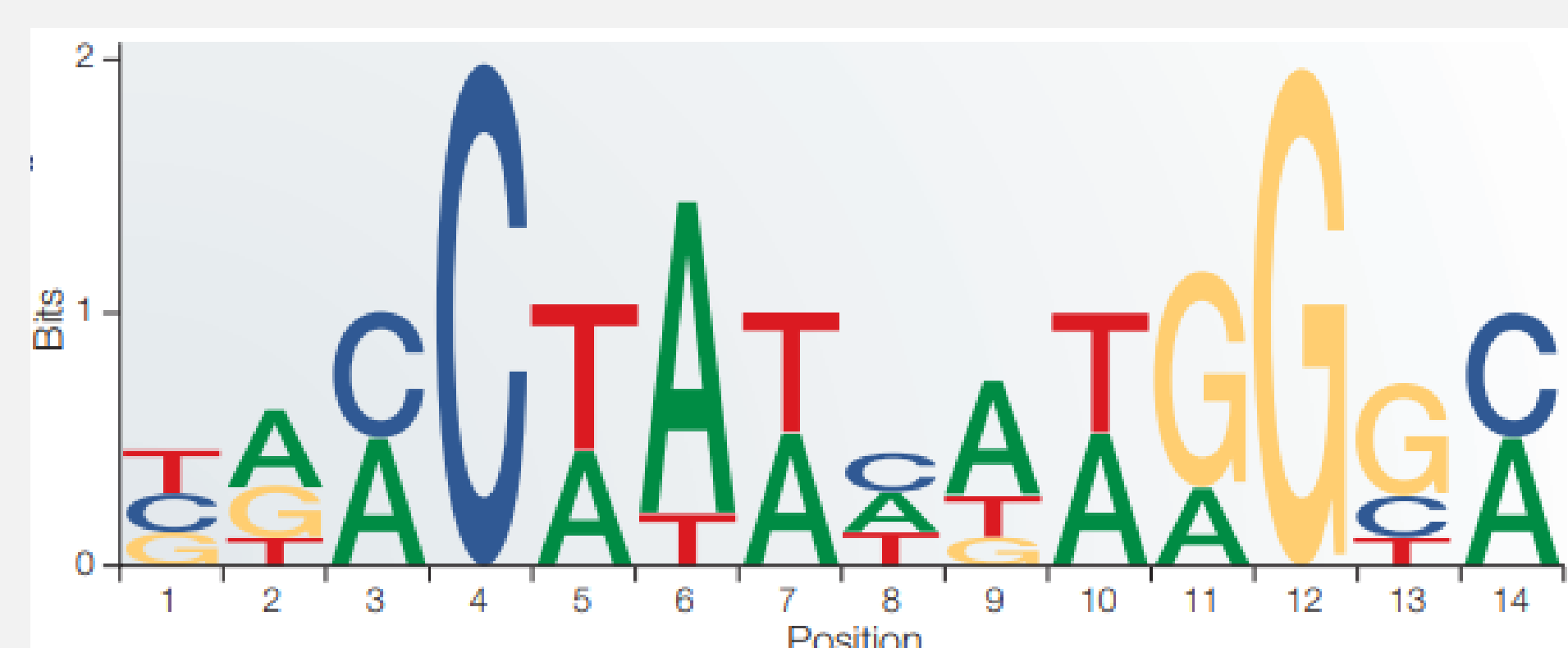
ChIP-seq experiments have shown that the same TF often binds at different loci depending on cell types[1]. The aim of this work is to identify the sequence features that determinate **the cell type specificities of TF binding**. We use for this a binary classification framework where given a specific TF and two cell types, the goal is to predict if a sequence is bound by the target TF in cell type 1 or 2. We use for this two sets of sequences (1000bp sequences centred on the target TF) derived from **ChIP-seq** experiments: the sequences bound only in cell type 1 and the sequences bound only in cell type 2.

**Three kinds of sequence features** are used to discriminate the two cell types:

- the nucleotide specificity of the binding site
- the nucleotide content around the binding site
- the presence and position of potential binding sites of other cooperative TFs

## Nucleotide specificity of the binding sites

The set of k-mers recognised by a TF are usually slight variations of a common motif. Traditionally, these k-mers are summarised in a probabilistic model known as a **Position Weight Matrix (PWM)**. PWMs of many TFs are available in dedicated database (JASPAR [2]).



TF: MEF2, image adapted from Wasserman and Sandelin April 2004[3]

Such motif can be used to scan a DNA sequence and to identify potential binding sites of the associated TF.

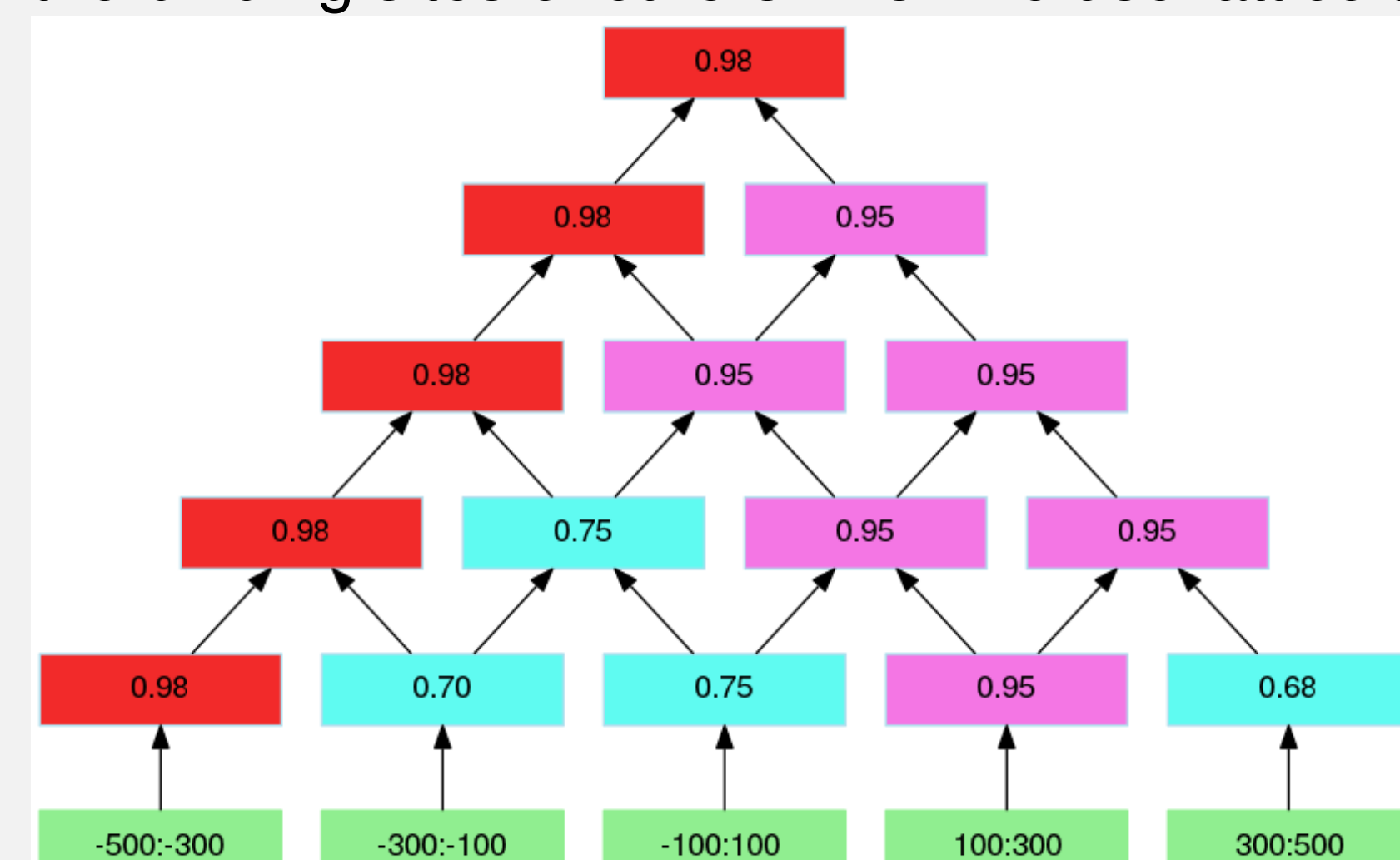
Here, rather than using directly the PWM of the target TF, we learn a new model based on a linear model that allows us to identify for each position the nucleotide differences between the two cell types. This model is called **"discriminative PWM"**.

## Nucleotide content around binding sites

We used the method DExTER[4] to capture the nucleotide specificity around the binding sites in the two cell types. Given two sets of sequences, DExTER identifies pairs of (k-mer, region) for which the frequency of the given k-mer in the given region is different between the two sets.

## Binding site positions of cooperative TFs

The third kind of features used to discriminate the two cell types is the position of the binding sites of others TFs. We use lattice to model this positional information.

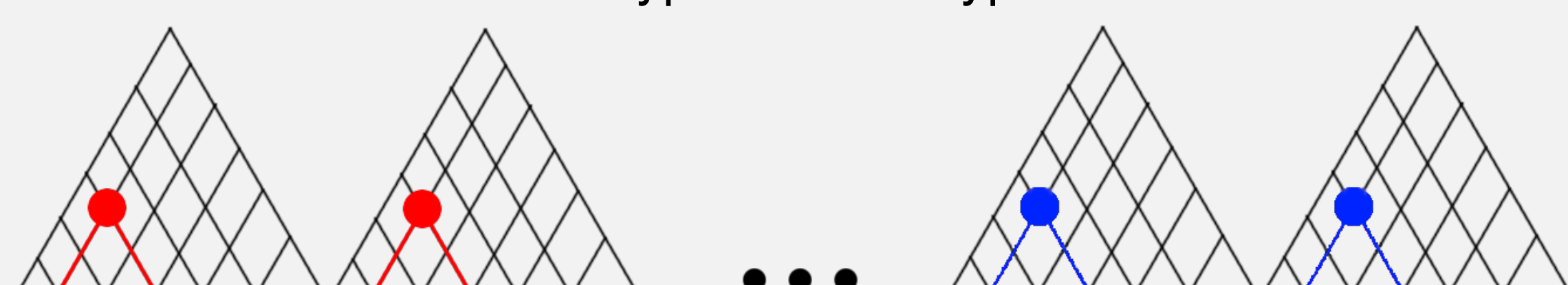


- Each node of the lattice is associated with a specific region of the sequence.
- This allows to quickly explore many different regions.

This lattice is used to compute the binding affinity of each PWM of the JASPAR database in each region of each sequence. Given a PWM, we then have a lattice associated with each sequence:

Each node of the lattice contains the binding affinity (computed with the PWM) of the sequence region associated with this node.

We have two sets of lattices: cell type 0 or cell type 1



- We then search for the region (lattice node) for which the differences of affinity between the two sets of lattices is the highest



## Global model

At this stage, we have three kinds of variables identified by three different approaches:

- The nucleotide specificities of the binding site of the target TF, identified with the discriminative PWM ( $S_i$ )
- The nucleotide content around the binding site of the target TF, identified by the method DExTER[4] ( $D_j$ )
- The positional information of cooperative TFs, identified using lattices ( $R_k$ )

We perform a logistic regression with L1 penalisation - (**LASSO**).

$$\ln \frac{\mathbb{P}(Y = 1 \mid X)}{1 - \mathbb{P}(Y = 1 \mid X)} = \beta_0 + \sum_{i=1}^s \beta_i S_i + \sum_{j=1}^d \gamma_j D_j + \sum_{k=1}^r \alpha_k R_k + \epsilon,$$

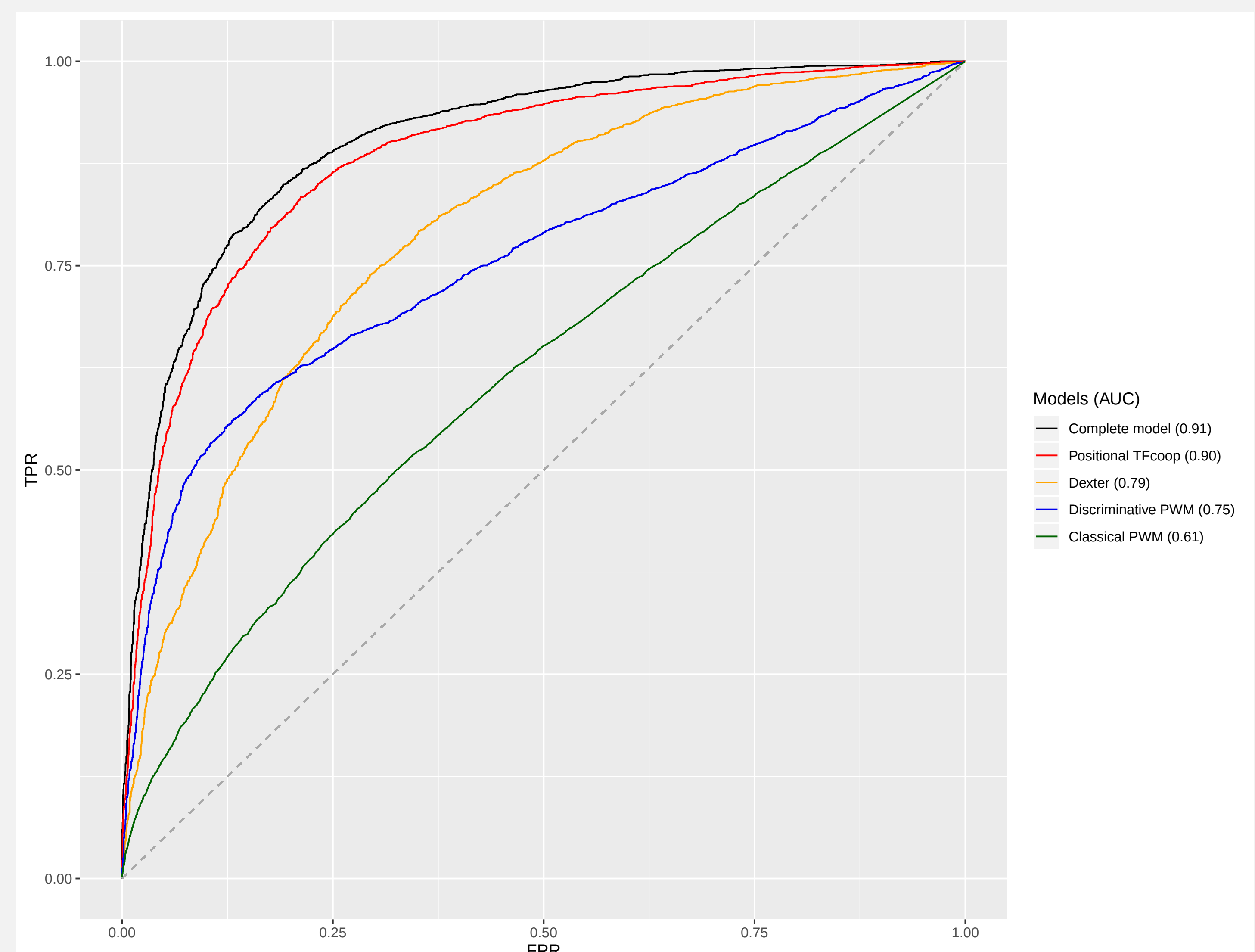
where  $X = (S_1, \dots, S_s, D_1, \dots, D_s, R_1, \dots, R_r)$  and  $\epsilon \sim N(0, \sigma^2)$

## Results

We apply our approach for discriminating the sequences bound by TF ESR1 in cell types MCF7 and T47D.

In this experiment, we use:

- 10380 sequences in each classes
- 14532 sequences (70%) in training to identify the three kind of features and learn  $(\beta, \gamma, \alpha)$  coefficients
- 6228 sequences (30%) in testing



## Conclusion

Those results show that the three types of information have good predictive performances. In this example, the nucleotide specificity of the binding site of the target TF and the positional information of all other cooperative TFs are the most important features to explain the differences of binding sites between the two cell types. We can also see that our discriminative PWM has better accuracy than the classical PWM for this problem.

We are now applying this approach to many other TFs and cell types to identify in each case, the sequences features responsible for the binding site differences between cell types.

## References

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